

```
-r          Bypass the normal routing and send directly to a
            host on an attached network
-s src_addr --source=src_addr    Use source src_addr for outgoing packets
-z sendwait --sendwait=sendwait Minimal time interval between probes (default 0).
                                If the value is more than 10, then it specifies a
                                number in milliseconds, else it is a number of
                                seconds (float point values allowed too)
-e --extensions                  Show ICMP extensions (if present), including MPLS
-A --as-path-lookups            Perform AS path lookups in routing registries and
                                print results directly after the corresponding
                                addresses
-M name --module=name          Use specified module (either builtin or external)
                                for traceroute operations. Most methods have
                                their shortcuts ('-I' means '-M icmp' etc.)
-O OPTS,... --options=OPTS,... Use module-specific option OPTS for the
                                traceroute module. Several OPTS allowed,
                                separated by commas. If OPTS is "help", print info
                                about available options
--sport=num                     Use source port num for outgoing packets. Implies
                                '-N 1'
--fwmark=num                    Set firewall mark for outgoing packets
-U --udp                        Use UDP to particular port for tracerouting
                                (instead of increasing the port per each probe),
                                default port is 53
-UL                             Use UDPLITE for tracerouting (default dest port
                                is 53)
-D --dccp                      Use DCCP Request for tracerouting (default port
                                is 33434)
-P prot --protocol=prot        Use raw packet of protocol prot for tracerouting
--mtu                           Discover MTU along the path being traced. Implies
                                '-F -N 1'
--back                          Guess the number of hops in the backward path and
                                print if it differs
-V --version                    Print version info and exit
--help                          Read this help and exit

Arguments:
+ host          The host to traceroute to
  packetlen    The full packet length (default is the length of an IP
               header plus 40). Can be ignored or increased to a minimal
               allowed value

[jeb@localhost ~]$
```

```
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=98 ttl=112 time=112 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=99 ttl=112 time=134 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=100 ttl=112 time=155 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=101 ttl=112 time=66.1 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=102 ttl=112 time=78.1 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=103 ttl=112 time=90.0 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=104 ttl=112 time=110 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=105 ttl=112 time=131 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=106 ttl=112 time=154 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=107 ttl=112 time=135 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=108 ttl=112 time=250 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=109 ttl=112 time=76.4 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=110 ttl=112 time=98.1 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=111 ttl=112 time=119 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=112 ttl=112 time=141 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=113 ttl=112 time=60.1 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=114 ttl=112 time=83.1 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=115 ttl=112 time=106 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=116 ttl=112 time=128 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=117 ttl=112 time=150 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=118 ttl=112 time=170 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=119 ttl=112 time=193 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=120 ttl=112 time=213 ms
64 bytes from bon07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=121 ttl=112 time=73.6 ms
^C
--- google.com ping statistics ---
121 packets transmitted, 121 received, 0% packet loss, time 121968ms
rtt min/avg/max/ndev = 24.262/92.859/315.927/49.513 ms
[jeb@localhost ~]$ ifconfig
enp6s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe2a:da7b prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:2a:da:7b txqueuelen 1000 (Ethernet)
    RX packets 828 bytes 75337 (73.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 882 bytes 78398 (76.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 19 bytes 2167 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 19 bytes 2167 (2.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
[jebin@localhost ~]$ mkdir JEBIN_241801103
[jebin@localhost ~]$ cd JEBIN_241801103
[jebin@localhost JEBIN_241801103]$ date
Wednesday 21 January 2026 08:33:51 AM IST
[jebin@localhost JEBIN_241801103]$ echo "This is Linux OS"
This is Linux OS
[jebin@localhost JEBIN_241801103]$ cal october 2006
October 2006
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

[jebin@localhost JEBIN_241801103]$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
33+228
261
3.3+22.8
26.1
quit
[jebin@localhost JEBIN_241801103]$ who
jebin    tty2          2026-01-21 08:31 (local)
[jebin@localhost JEBIN_241801103]$ who am i
[jebin@localhost JEBIN_241801103]$ id
uid=1000(jebin) gid=1000(jebin) groups=1000(jebin),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[jebin@localhost JEBIN_241801103]$ tty
/dev/pts/0
[jebin@localhost JEBIN_241801103]$ man touch os.txt
No manual entry for os.txt
[jebin@localhost JEBIN_241801103]$ ps
  PID TTY          TIME CMD
  2648 pts/0    00:00:00 bash
  2858 pts/0    00:00:00 ps
[jebin@localhost JEBIN_241801103]$ uname
Linux
[jebin@localhost JEBIN_241801103]$ uname -m
x86_64
[jebin@localhost JEBIN_241801103]$ uname -n
localhost.localdomain
[jebin@localhost JEBIN_241801103]$ uname -a
Linux localhost.localdomain 5.14.0-061.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jan 16 09:19:22 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
[jebin@localhost JEBIN_241801103]$ mkdir rec
```

```
7 root      0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-netns
9 root      0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/0:0H-events_highpri
11 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-mn_percpu_wq
12 root    20  8  0  0  0 1 0.0 0.0 0:04.01 worker/0:1-events_unbound
13 root    20  0  0  0  0 1 0.0 0.0 0:00.00 rcu_tasks_kthre
14 root    20  0  0  0  0 1 0.0 0.0 0:00.00 rcu_tasks_rude_
15 root    20  0  0  0  0 1 0.0 0.0 0:00.00 rcu_tasks_trace
16 root    20  0  0  0  0 5 0.0 0.0 0:00.04 ksoftirqd/0
17 root    20  0  0  0  0 1 0.0 0.0 0:04.76 rcu_preempt
18 root    20  0  0  0  0 5 0.0 0.0 0:00.00 rcu_exp_par_0p_
19 root    20  0  0  0  0 5 0.0 0.0 0:00.10 rcu_exp_gp_kthr
20 root     rt  0  0  0  0 5 0.0 0.0 0:00.23 migration/0
21 root   -51  0  0  0  0 5 0.0 0.0 0:00.00 idle_inject/0
23 root    20  0  0  0  0 5 0.0 0.0 0:00.00 cpuhp/0
24 root    20  0  0  0  0 5 0.0 0.0 0:00.00 cpuhp/1
25 root   -51  0  0  0  0 5 0.0 0.0 0:00.00 idle_inject/1
26 root     rt  0  0  0  0 5 0.0 0.0 0:00.26 migration/1
27 root    20  0  0  0  0 5 0.0 0.0 0:02.12 ksoftirqd/1
29 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/1:0H-kblockd
31 root    20  0  0  0  0 5 0.0 0.0 0:00.07 kdevtmpfs
32 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-inet_frag_wq
33 root    20  0  0  0  0 5 0.0 0.0 0:00.01 kauditd
34 root    20  0  0  0  0 5 0.0 0.0 0:00.02 khungtaskd
36 root    20  0  0  0  0 5 0.0 0.0 0:00.00 oom_reaper
37 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-writeback
38 root    20  0  0  0  0 5 0.0 0.0 0:00.67 kcompactd0
39 root    25  5  0  0  0 5 0.0 0.0 0:00.00 ksmd
40 root    30 10  0  0  0 5 0.0 0.0 0:03.51 khugepaged
41 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-cryptd
42 root     0 -20  0  0  0 1 0.0 0.0 0:00.00 worker/8-kblockd
[jeb@localhost rec]$ ps
  PID TTY          TIME CMD
 2648 pts/0    00:00:00 bash
 3341 pts/0    00:00:00 ps
[jeb@localhost rec]$ vmstat
procs-----memory-----swap-----io-----system-----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st
 5 0  0 349500 2860 659472 0 0 139 5 155 186 2 4 95 0 0
[jeb@localhost rec]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        821M   0  821M   0% /dev
tmpfs           850M   0  850M   0% /dev/shm
tmpfs           342M  5.5M  337M   2% /run
/dev/mapper/c3-root 27G  5.0G  22G  19% /
/dev/sda1       560M  439M  121M  78% /boot
tmpfs          171M 104K  171M   1% /run/user/1000
/dev/sr0        51M   51M   0 100% /run/media/jeb/VBox_GAs_7.8.12
```

```
[jebini@localhost test]$ cd .. || echo "Directory changed"
[jebini@localhost rec]$ ls
f1.txt  os.txt  test
[jebini@localhost rec]$ cd rec
bash: cd: rec: No such file or directory
[jebini@localhost rec]$ ls
f1.txt  os.txt  test
[jebini@localhost rec]$ head f1.txt
I am Ironman
[jebini@localhost rec]$ head -5 f1.txt
I am Ironman
[jebini@localhost rec]$ ls | more
f1.txt
os.txt
test
[jebini@localhost rec]$ grep "n" f1.txt
I am Ironman
[jebini@localhost rec]$ nl os.txt
[jebini@localhost rec]$ sort f1.txt
I am Ironman
[jebini@localhost rec]$ sort -r f1.txt
I am Ironman
[jebini@localhost rec]$ nl f1.txt
1 I am Ironman
[jebini@localhost rec]$ cut -c 1-8 f1.txt
I am Iron
[jebini@localhost rec]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.7Gi          903Mi        332Mi         19Mi        646Mi        804Mi
Swap:          2.8Gi           0B          2.8Gi
[jebini@localhost rec]$ top

top - 13:06:05 up 42 min, 1 user, load average: 0.09, 0.15, 0.17
Tasks: 195 total, 1 running, 194 sleeping, 0 stopped, 0 zombie
%cpu(s):  0.2 us,  0.7 sy,  0.0 ni, 98.6 id,  0.2 wa,  0.0 hi,  0.3 si,  0.0 st
MiB Mem : 1797.4 total, 332.6 free, 982.9 used, 646.8 buff/cache
MiB Swap: 2896.0 total, 2896.0 free,  0.0 used, 804.5 avail Mem

  PID USER      PR  PT  VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 2565 root        20   0   640996 25480 20780 S   0.3   1.5   0:01.24 fwupd
 2630 jebini      20   0   774432 52396 40924 S   0.3   3.0   0:21.18 gnome-terminal-
3333 jebini      20   0   225920 4412   3596 R   0.3   0.3   0:01.94 top
    1 root        20   0   174768 17224 11180 S   0.0   1.0   0:05.90 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.11 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_
    4 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
```

```
[jebin@localhost JEBIN_241801103]$ mkdir rec
[jebin@localhost JEBIN_241801103]$ cd rec
[jebin@localhost rec]$
[jebin@localhost rec]$ rmdir rec
rmdir: failed to remove 'rec': No such file or directory
[jebin@localhost rec]$ ls
[jebin@localhost rec]$ touch os.txt
[jebin@localhost rec]$ vi os.txt
[jebin@localhost rec]$ cat os.txt
[jebin@localhost rec]$ touch os2.txt
[jebin@localhost rec]$ cp os.txt os2.txt
[jebin@localhost rec]$ cat os2.txt
[jebin@localhost rec]$ rm os2.txt
[jebin@localhost rec]$ ls
os.txt
[jebin@localhost rec]$ mkdir rec
[jebin@localhost rec]$ mv os.txt rec
[jebin@localhost rec]$ cd rec
[jebin@localhost rec]$ ls
os.txt
[jebin@localhost rec]$ file os.txt
os.txt: empty
[jebin@localhost rec]$ touch f1.txt
[jebin@localhost rec]$ vi f1.txt
[jebin@localhost rec]$ cat f1.txt
I am Ironman
[jebin@localhost rec]$ file f1.txt
f1.txt: ASCII text
[jebin@localhost rec]$ wc f1.txt
 1  3 13 f1.txt
[jebin@localhost rec]$ ls
f1.txt  os.txt
[jebin@localhost rec]$ who | wc -l
1
[jebin@localhost rec]$ who | who | wc
      1      5      47
[jebin@localhost rec]$ who | date
Wednesday 21 January 2026 12:58:18 PM IST
[jebin@localhost rec]$ ls r++
ls: cannot access 'r++': No such file or directory
[jebin@localhost rec]$ f1.txt
bash: f1.txt: command not found...
[jebin@localhost rec]$ who ; date
jebin      tty2      2026-01-21 08:31 (local)
Wednesday 21 January 2026 12:59:38 PM IST
[jebin@localhost rec]$ mkdir test && cd test
[jebin@localhost test]$ ls cd ... || echo "Directory changed"
```

```
[jeb@localhost ~]$ cd JEBIN_241801103
[jeb@localhost JEBIN_241801103]$ date
Wednesday 21 January 2026 06:33:51 AM IST
[jeb@localhost JEBIN_241801103]$ echo "This is Linux OS"
This is Linux OS
[jeb@localhost JEBIN_241801103]$ cal october 2006
October 2006
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

[jeb@localhost JEBIN_241801103]$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
33+228
261
1.3+22.8
26.1
quit
[jeb@localhost JEBIN_241801103]$ who
jeb   tty2      2026-01-21 06:31 (local)
[jeb@localhost JEBIN_241801103]$ who am i
jeb   10      1000(jeb)  groups=1000(jeb),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[jeb@localhost JEBIN_241801103]$ tty
/dev/pts/0
[jeb@localhost JEBIN_241801103]$ man touch os.txt
No manual entry for os.txt
[jeb@localhost JEBIN_241801103]$ ps
  PID TTY          TIME CMD
 2608 pts/0    00:00:00 bash
 2808 pts/0    00:00:00 ps
[jeb@localhost JEBIN_241801103]$ uname
Linux
[jeb@localhost JEBIN_241801103]$ uname -m
x86_64
[jeb@localhost JEBIN_241801103]$ uname -n
localhost.localdomain
[jeb@localhost JEBIN_241801103]$ uname -a
Linux localhost.localdomain 5.14.0-661.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jan 16 09:19:22 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
[jeb@localhost JEBIN_241801103]$
```